

Windows PowerShell

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<https://aka.ms/PSWindows>

```
PS C:\Users\gubba> mongosh --version
```

```
2.13.0
```

```
PS C:\Users\gubba> mongosh
```

```
Current Mongosh Log ID: 6ac7612c4c60b3c653292c9f
```

```
Connecting to:
```

```
mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.13.0
```

```
Using MongoDB:      8.3.9
```

```
Using Mongosh:      2.13.0
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

```
-----
```

The server generated these startup warnings when booting

```
2026-10-06T10:56:48.948+05:30: Access control is not enabled for the database.
Read and write access to data and configuration is unrestricted
```

```
-----
```

```
test> mongosh
```

```
ReferenceError: mongosh is not defined
```

```
test> use collegeDB
```

```
switched to db collegeDB
```

```
collegeDB> db.createCollection("students")
```

```
{ ok: 1 }
```

```
collegeDB> db.students.insertMany([  
  | {rollno: 101, name: "Arun", department: "CSE", year: 3, marks: 85},  
  | {rollno: 102, name: "Bala", department: "ECE", year: 2, marks: 78},  
  | {rollno: 103, name: "Charan", department: "CSE", year: 3, marks: 92},  
  | {rollno: 104, name: "Deepa", department: "EEE", year: 1, marks: 69},  
  | {rollno: 105, name: "Esha", department: "CSE", year: 2, marks: 88},  
  | {rollno: 106, name: "Farhan", department: "ECE", year: 3, marks: 76},  
  | {rollno: 107, name: "Gokul", department: "CSE", year: 3, marks: 95},  
  | {rollno: 108, name: "Harini", department: "IT", year: 2, marks: 81},  
  | {rollno: 109, name: "Ishaan", department: "CSE", year: 1, marks: 73},  
  | {rollno: 110, name: "Jaya", department: "ECE", year: 3, marks: 90}  
  | ])
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6ac761904c60b3c653292ca0'),  
    '1': ObjectId('6ac761904c60b3c653292ca1'),  
    '2': ObjectId('6ac761904c60b3c653292ca2'),  
    '3': ObjectId('6ac761904c60b3c653292ca3'),  
    '4': ObjectId('6ac761904c60b3c653292ca4'),  
    '5': ObjectId('6ac761904c60b3c653292ca5'),  
    '6': ObjectId('6ac761904c60b3c653292ca6'),  
    '7': ObjectId('6ac761904c60b3c653292ca7'),  
    '8': ObjectId('6ac761904c60b3c653292ca8'),  
    '9': ObjectId('6ac761904c60b3c653292ca9')  
  }  
}
```

```
collegeDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac761904c60b3c653292ca0'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca1'),
    rollno: 102,
    name: 'Bala',
    department: 'ECE',
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca2'),
    rollno: 103,
    name: 'Charan',
    department: 'CSE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca3'),
```

```
rollno: 104,  
name: 'Deepa',  
department: 'EEE',  
year: 1,  
marks: 69  
,  
{  
  _id: ObjectId('6ac761904c60b3c653292ca4'),  
  rollno: 105,  
  name: 'Esha',  
  department: 'CSE',  
  year: 2,  
  marks: 88  
,  
{  
  _id: ObjectId('6ac761904c60b3c653292ca5'),  
  rollno: 106,  
  name: 'Farhan',  
  department: 'ECE',  
  year: 3,  
  marks: 76  
,  
{  
  _id: ObjectId('6ac761904c60b3c653292ca6'),  
  rollno: 107,  
  name: 'Gokul',  
  department: 'CSE',  
  year: 3,
```

```
marks: 95
},
{
  _id: ObjectId('6ac761904c60b3c653292ca7'),
  rollno: 108,
  name: 'Harini',
  department: 'IT',
  year: 2,
  marks: 81
},
{
  _id: ObjectId('6ac761904c60b3c653292ca8'),
  rollno: 109,
  name: 'Ishaan',
  department: 'CSE',
  year: 1,
  marks: 73
},
{
  _id: ObjectId('6ac761904c60b3c653292ca9'),
  rollno: 110,
  name: 'Jaya',
  department: 'ECE',
  year: 3,
  marks: 90
}
]
```

```
collegeDB> db.students.find()
```

```
[
  {
    _id: ObjectId('6ac761904c60b3c653292ca0'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca1'),
    rollno: 102,
    name: 'Bala',
    department: 'ECE',
    year: 2,
    marks: 78
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca2'),
    rollno: 103,
    name: 'Charan',
    department: 'CSE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca3'),
    rollno: 104,
```

```
name: 'Deepa',
department: 'EEE',
year: 1,
marks: 69
},
{
  _id: ObjectId('6ac761904c60b3c653292ca4'),
  rollno: 105,
  name: 'Esha',
  department: 'CSE',
  year: 2,
  marks: 88
},
{
  _id: ObjectId('6ac761904c60b3c653292ca5'),
  rollno: 106,
  name: 'Farhan',
  department: 'ECE',
  year: 3,
  marks: 76
},
{
  _id: ObjectId('6ac761904c60b3c653292ca6'),
  rollno: 107,
  name: 'Gokul',
  department: 'CSE',
  year: 3,
  marks: 95
```

```
},
{
  _id: ObjectId('6ac761904c60b3c653292ca7'),
  rollno: 108,
  name: 'Harini',
  department: 'IT',
  year: 2,
  marks: 81
},
{
  _id: ObjectId('6ac761904c60b3c653292ca8'),
  rollno: 109,
  name: 'Ishaan',
  department: 'CSE',
  year: 1,
  marks: 73
},
{
  _id: ObjectId('6ac761904c60b3c653292ca9'),
  rollno: 110,
  name: 'Jaya',
  department: 'ECE',
  year: 3,
  marks: 90
}
]
collegeDB> db.students.find({department: "CSE"})
[
```

```
{
  _id: ObjectId('6ac761904c60b3c653292ca0'),
  rollno: 101,
  name: 'Arun',
  department: 'CSE',
  year: 3,
  marks: 85
},
{
  _id: ObjectId('6ac761904c60b3c653292ca2'),
  rollno: 103,
  name: 'Charan',
  department: 'CSE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac761904c60b3c653292ca4'),
  rollno: 105,
  name: 'Esha',
  department: 'CSE',
  year: 2,
  marks: 88
},
{
  _id: ObjectId('6ac761904c60b3c653292ca6'),
  rollno: 107,
  name: 'Gokul',
```

```
    department: 'CSE',
    year: 3,
    marks: 95
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca8'),
    rollno: 109,
    name: 'Ishaan',
    department: 'CSE',
    year: 1,
    marks: 73
  }
]
collegeDB> db.students.find({marks: {$gt: 80}})
[
  {
    _id: ObjectId('6ac761904c60b3c653292ca0'),
    rollno: 101,
    name: 'Arun',
    department: 'CSE',
    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca2'),
    rollno: 103,
    name: 'Charan',
    department: 'CSE',
```

```
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca4'),
    rollno: 105,
    name: 'Esha',
    department: 'CSE',
    year: 2,
    marks: 88
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca6'),
    rollno: 107,
    name: 'Gokul',
    department: 'CSE',
    year: 3,
    marks: 95
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca7'),
    rollno: 108,
    name: 'Harini',
    department: 'IT',
    year: 2,
    marks: 81
  },
  {
```

```
  _id: ObjectId('6ac761904c60b3c653292ca9'),  
  rollno: 110,  
  name: 'Jaya',  
  department: 'ECE',  
  year: 3,  
  marks: 90  
}  
]
```

```
collegeDB> db.students.find({}, {_id: 0, name: 1, marks: 1})
```

```
[  
  { name: 'Arun', marks: 85 },  
  { name: 'Bala', marks: 78 },  
  { name: 'Charan', marks: 92 },  
  { name: 'Deepa', marks: 69 },  
  { name: 'Esha', marks: 88 },  
  { name: 'Farhan', marks: 76 },  
  { name: 'Gokul', marks: 95 },  
  { name: 'Harini', marks: 81 },  
  { name: 'Ishaan', marks: 73 },  
  { name: 'Jaya', marks: 90 }  
]
```

```
collegeDB> db.students.find({department: "CSE", year: 3})
```

```
[  
  {  
    _id: ObjectId('6ac761904c60b3c653292ca0'),  
    rollno: 101,  
    name: 'Arun',  
    department: 'CSE',
```

```

    year: 3,
    marks: 85
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca2'),
    rollno: 103,
    name: 'Charan',
    department: 'CSE',
    year: 3,
    marks: 92
  },
  {
    _id: ObjectId('6ac761904c60b3c653292ca6'),
    rollno: 107,
    name: 'Gokul',
    department: 'CSE',
    year: 3,
    marks: 95
  }
]
collegeDB> db.students.find({
| $or: [
|   {department: "CSE"},
|   {department: "ECE"}
| ]
| })
[
  {

```

```
_id: ObjectId('6ac761904c60b3c653292ca0'),
rollno: 101,
name: 'Arun',
department: 'CSE',
year: 3,
marks: 85
},
{
  _id: ObjectId('6ac761904c60b3c653292ca1'),
  rollno: 102,
  name: 'Bala',
  department: 'ECE',
  year: 2,
  marks: 78
},
{
  _id: ObjectId('6ac761904c60b3c653292ca2'),
  rollno: 103,
  name: 'Charan',
  department: 'CSE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac761904c60b3c653292ca4'),
  rollno: 105,
  name: 'Esha',
  department: 'CSE',
```

```
year: 2,
marks: 88
},
{
  _id: ObjectId('6ac761904c60b3c653292ca5'),
  rollno: 106,
  name: 'Farhan',
  department: 'ECE',
  year: 3,
  marks: 76
},
{
  _id: ObjectId('6ac761904c60b3c653292ca6'),
  rollno: 107,
  name: 'Gokul',
  department: 'CSE',
  year: 3,
  marks: 95
},
{
  _id: ObjectId('6ac761904c60b3c653292ca8'),
  rollno: 109,
  name: 'Ishaan',
  department: 'CSE',
  year: 1,
  marks: 73
},
{
```

```
_id: ObjectId('6ac761904c60b3c653292ca9'),  
rollno: 110,  
name: 'Jaya',  
department: 'ECE',  
year: 3,  
marks: 90  
}  
]
```

```
collegeDB> db.students.find().sort({marks: 1})
```

```
[  
{  
  _id: ObjectId('6ac761904c60b3c653292ca3'),  
  rollno: 104,  
  name: 'Deepa',  
  department: 'EEE',  
  year: 1,  
  marks: 69  
},  
{  
  _id: ObjectId('6ac761904c60b3c653292ca8'),  
  rollno: 109,  
  name: 'Ishaan',  
  department: 'CSE',  
  year: 1,  
  marks: 73  
},  
{  
  _id: ObjectId('6ac761904c60b3c653292ca5'),
```

```
rollno: 106,  
name: 'Farhan',  
department: 'ECE',  
year: 3,  
marks: 76  
,  
{  
  _id: ObjectId('6ac761904c60b3c653292ca1'),  
  rollno: 102,  
  name: 'Bala',  
  department: 'ECE',  
  year: 2,  
  marks: 78  
},  
{  
  _id: ObjectId('6ac761904c60b3c653292ca7'),  
  rollno: 108,  
  name: 'Harini',  
  department: 'IT',  
  year: 2,  
  marks: 81  
},  
{  
  _id: ObjectId('6ac761904c60b3c653292ca0'),  
  rollno: 101,  
  name: 'Arun',  
  department: 'CSE',  
  year: 3,
```

```
marks: 85
},
{
  _id: ObjectId('6ac761904c60b3c653292ca4'),
  rollno: 105,
  name: 'Esha',
  department: 'CSE',
  year: 2,
  marks: 88
},
{
  _id: ObjectId('6ac761904c60b3c653292ca9'),
  rollno: 110,
  name: 'Jaya',
  department: 'ECE',
  year: 3,
  marks: 90
},
{
  _id: ObjectId('6ac761904c60b3c653292ca2'),
  rollno: 103,
  name: 'Charan',
  department: 'CSE',
  year: 3,
  marks: 92
},
{
  _id: ObjectId('6ac761904c60b3c653292ca6'),
```

```
rollno: 107,  
name: 'Gokul',  
department: 'CSE',  
year: 3,  
marks: 95  
}  
]  
collegeDB> db.students.find().sort({marks: -1}).limit(3)  
[  
  {  
    _id: ObjectId('6ac761904c60b3c653292ca6'),  
    rollno: 107,  
    name: 'Gokul',  
    department: 'CSE',  
    year: 3,  
    marks: 95  
  },  
  {  
    _id: ObjectId('6ac761904c60b3c653292ca2'),  
    rollno: 103,  
    name: 'Charan',  
    department: 'CSE',  
    year: 3,  
    marks: 92  
  },  
  {  
    _id: ObjectId('6ac761904c60b3c653292ca9'),  
    rollno: 110,
```

```

    name: 'Jaya',
    department: 'ECE',
    year: 3,
    marks: 90
  }
]
collegeDB> db.students.countDocuments({department: "CSE"})
5
collegeDB> db.students.updateOne(
| {rollno: 101},
| {$set: {marks: 90}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
collegeDB> db.students.updateMany(
| {department: "CSE"},
| {$set: {status: "Active"}}
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,

```

```

    upsertedCount: 0
  }
collegeDB> db.students.deleteOne({rollno: 110})
{ acknowledged: true, deletedCount: 1 }
collegeDB> db.students.distinct("department")
['CSE', 'ECE', 'EEE', 'IT' ]
collegeDB> db.students.aggregate([
| {
|   $group: {
|     _id: "$department",
|     averageMarks: {$avg: "$marks"}
|   }
| }
|])
[
  { _id: 'IT', averageMarks: 81 },
  { _id: 'ECE', averageMarks: 77 },
  { _id: 'CSE', averageMarks: 87.6 },
  { _id: 'EEE', averageMarks: 69 }
]
collegeDB> db.students.aggregate([
| {
|   $group: {
|     _id: "$department",
|     maximumMarks: {$max: "$marks"}
|   }
| }
|])

```

```
[  
  { _id: 'ECE', maximumMarks: 78 },  
  { _id: 'CSE', maximumMarks: 95 },  
  { _id: 'EEE', maximumMarks: 69 },  
  { _id: 'IT', maximumMarks: 81 }  
]
```